RAAGUL GANANATHAN

SOFTWARE ENGINEER

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@RaagulGananathan | @RaagulGananathan13

A motivated final-year BSc Information Technology undergraduate at Sri Lanka Institute of Information Technology with a strong foundation in software development, database management, and systems analysis. Experienced in full-stack web development, AI/ML research, and contributing to production-deployed applications. Passionate about leveraging technology to solve real-world problems and seeking opportunities to grow within a professional IT environment.

AREA OF EXPERTISE

React.js	Express.js	JAVA	Python	AWS (EC2, S3)	Power BI	SAP
SQL	Node.js	Spring Boot	Fast API	Docker	Jupyter Notebook	ERP Domain

WORK EXPERIENCE

Software Engineer (Part Time) **Aug 2025 - Present**
Fexten Solutions (Pvt) Ltd

- Resolved production bugs across live ERP modules ensuring system stability
- Designed and implemented backend database schemas to support business operations
- Developed server-side logic and frontend UI components across POS, Inventory Management, HCM/Payroll, and Accounting & Finance modules
- Wrote and maintained test scripts to validate module functionality and ensure code quality
- Gained hands-on ERP domain knowledge across core business operations

Intern Software Engineer **Oct 2025 - Present**
Techorin

- Contributed to the development of the Genius Bees Educational Platform as an intern, working across frontend using JavaScript. Responsibilities included building UI pages, developing server-side logic, designing database schemas, and contributing to the admin panel for content and user management.

Intern Full Stack Developer **Feb 2025 - Sept 2025**
Sri Lanka Telecom PLC

- **NCPA Complaint Portal Revamp** — Revamped the NCPA complaint portal using the MERN stack, developing frontend components, backend modules, and contributing to BRD preparation.
- **Digital Identity Hub (MYSLT)** — Built backend RESTful APIs following TM Forum standards, Contributed to biometric login, Google Passkeys, and multi-device session tracking features.

EDUCATION

SRI LANKA INSTITUTE OF INFORMATION TECHNOLOGY **Feb 2023 - Nov 2026**
BSc (Hons) in Information Technology

- Deans List (Y3S2)

Primary & Secondary Education **Jan 2008 - July 2021**
Hindu College Colombo

PROJECTS & CONTRIBUTIONS

FedMed — Federated MultiModal Medical AI **RESEARCH · IN PROGRESS**

Privacy-preserving federated learning system for multi-disease diagnosis using 5-channel multi-modal inputs (chest X-rays, lab values, clinical notes). Implements custom DW-FedAvg aggregation with disease-aware fusion, achieving 0.89 AUROC across 2 hospitals without sharing patient data.

Tech Stack: Python, PyTorch, Federated Learning, DenseNet-121, BioClinicalBERT

Candy Shop E-commerce (UK Client) **PROFESSIONAL · IN PROGRESS**

Full-stack e-commerce platform for a UK-based candy retailer with product catalog, user authentication, shopping cart, checkout, and admin dashboard. Designed for production deployment with responsive mobile-first UI.

Tech Stack: React 19, Tailwind CSS, Node.js, Express.js, MySQL, Firebase, Selenium

ShopsTime - UK Staff Wages Portal **DEPLOYED · LIVE**

Production-deployed supermarket staff management system for a UK-based client. Includes employee management, payroll exports, attendance tracking, and automated Excel report generation.

Tech Stack: React 19, Tailwind CSS, Node.js, Express.js, MySQL, AWS

Brain Tumor Detection System **ACADEMIC**

AI-powered brain tumor classification achieving 99.39% accuracy using an ensemble of 4 deep learning models. Features 6 XAI visualization methods, batch processing, and AI-generated medical reports via Gemini 2.0 Flash.

Tech Stack: PyTorch, Python, Flask, face-api.js, XAI, MongoDB

Car Service Management System with AI Damage Detection and E-Commerce Integration **ACADEMIC**

The Car Service Management System is a web platform using AI to detect vehicle damage, manage service appointments, and support e-commerce for parts. Built with the MERN stack, it features real-time tracking, an AI chatbot for assistance, and WhatsApp notifications. It improved service speed by 30% and received 85% user satisfaction in pilot tests.

Tech Stack: React 19, Tailwind CSS, Node.js, Express.js, MySQL, Firebase, Selenium

Skill-Sharing & Learning Platform **ACADEMIC**

Developed a coding focused skill-sharing platform using Spring Boot, allowing users to post images/videos, track learning progress, and share structured learning plans. Implemented interactive features such as likes, editable/deletable comments, and follower based content feeds. Integrated OAuth

Tech Stack: JAVA, SpringBoot, React 19, Tailwind CSS, Node.js, Express.js, PostgreSQL

REFERENCES

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